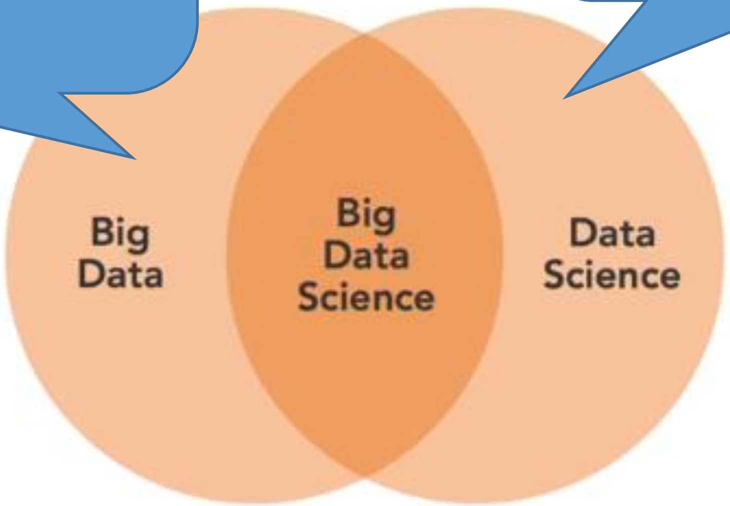
Introduction to Data Science:

Difference between
Data Science, Data Analysis, Data Analytics, Data Mining

Big Data Science?

Concerned about collection, storage, aggregation, communication of LARGE pools of data (which has the 4Vs characteristics).
May not need the algorithm process for analysis.

Traditional data science deals mostly structured data, smaller volume of data, data that aren't updated frequently.

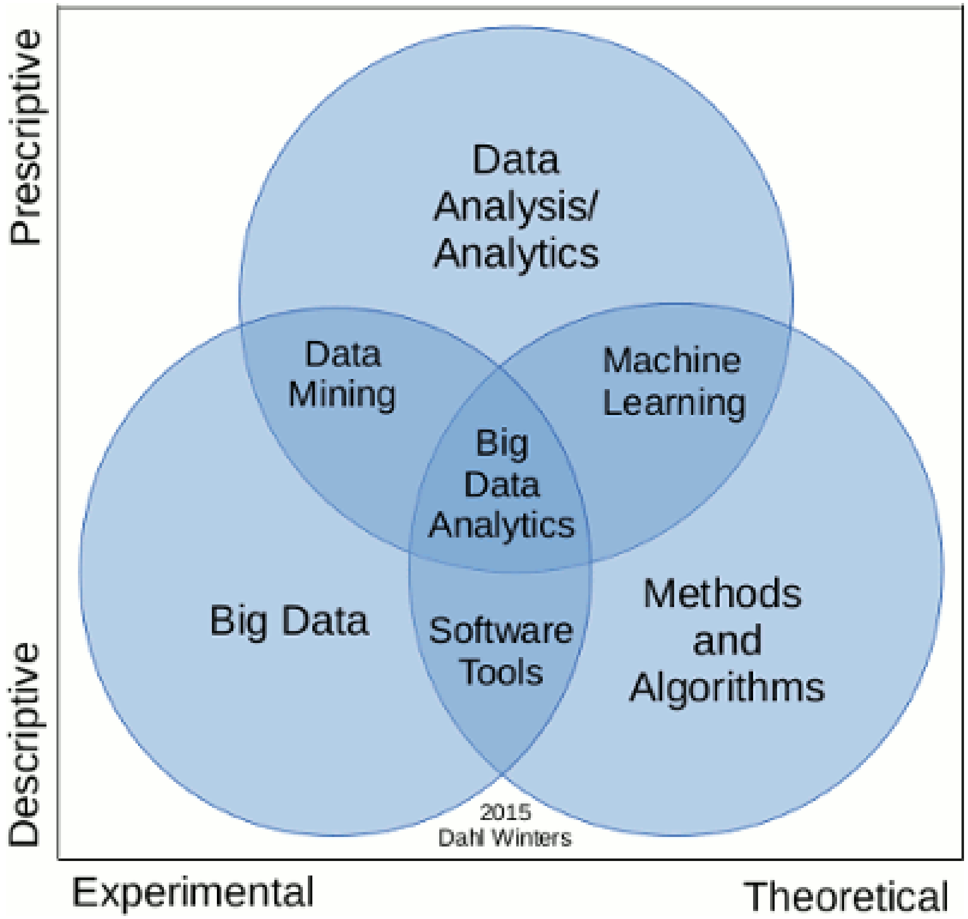


- **Combines** the challenges and the skills of the two domains. E.g.
  - ➢Techniques to analyse data need to take care of the 4Vs of Big Data.
  - ➢Will use mathematical/statistical methods on Big Data.

Data Mining Vs Data Analysis Vs Data Analytics

Why is it happening?

What had happened?



Purely based on data

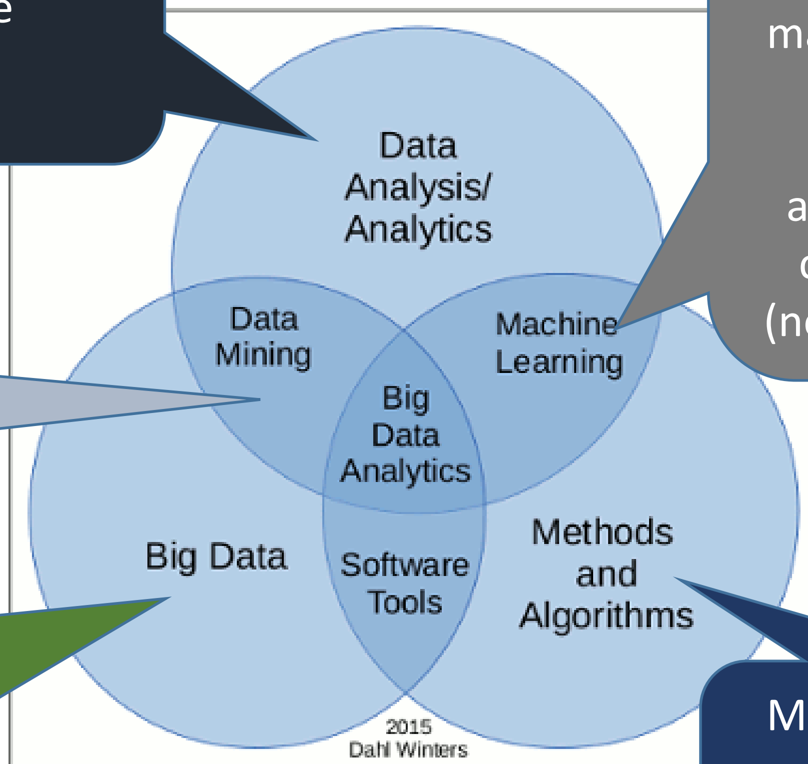
Purely based on mathematical models

Comparison at a glance

Able to tell based on why it happening and make plans/decisions

Try to discover pattern/trends but may not know the reason(s) why

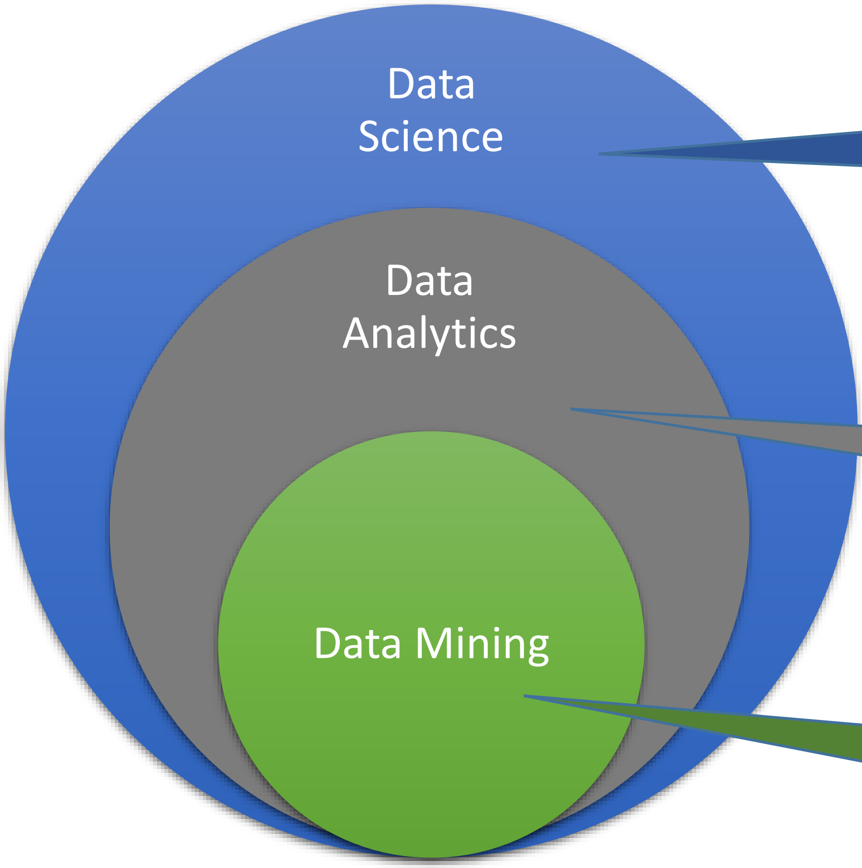
No modelling (which needs equation). Just the usual 3Ms, graphing, etc. May not even discover pattern yet.



Models are derived from mixture of solid mathematical background & tuning (thus experimental). Model alone WITHOUT context cannot guide decisions (not very prescriptive yet).

Math equations. Without data, cannot enable decisions (so NOT prescriptive)

Hierarchical Comparison



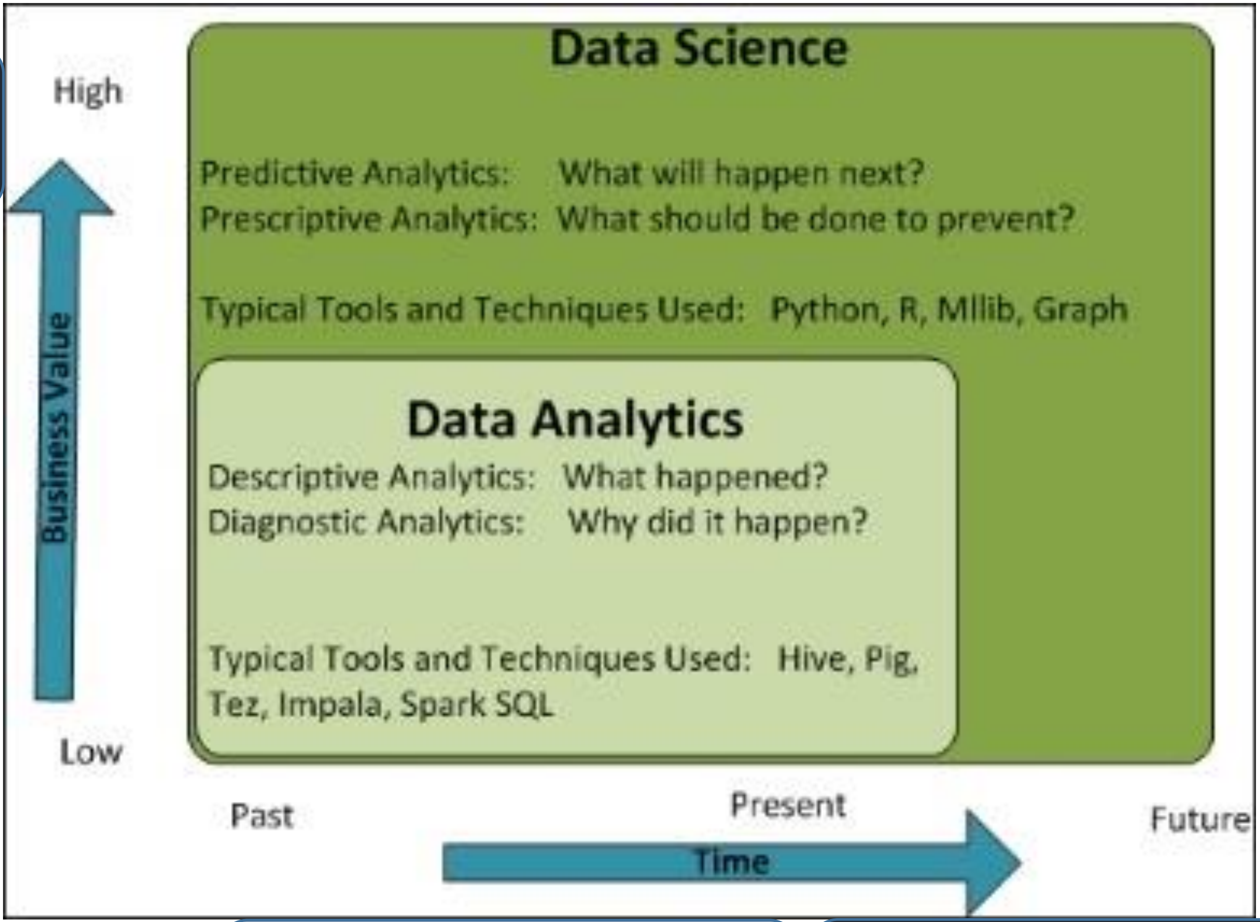
Highest level: not only predictive (prepared for the future) but prescriptive (know what to do) too.

Mid level: use of statistics and tools to enable algorithms/modelling to find insights. Already know what to look for.

Low level: don’t know what is the problem yet, just discover pattern, make messy data into consistent and tidy ones, to enable the task above.

Data Science looks beyond what is available

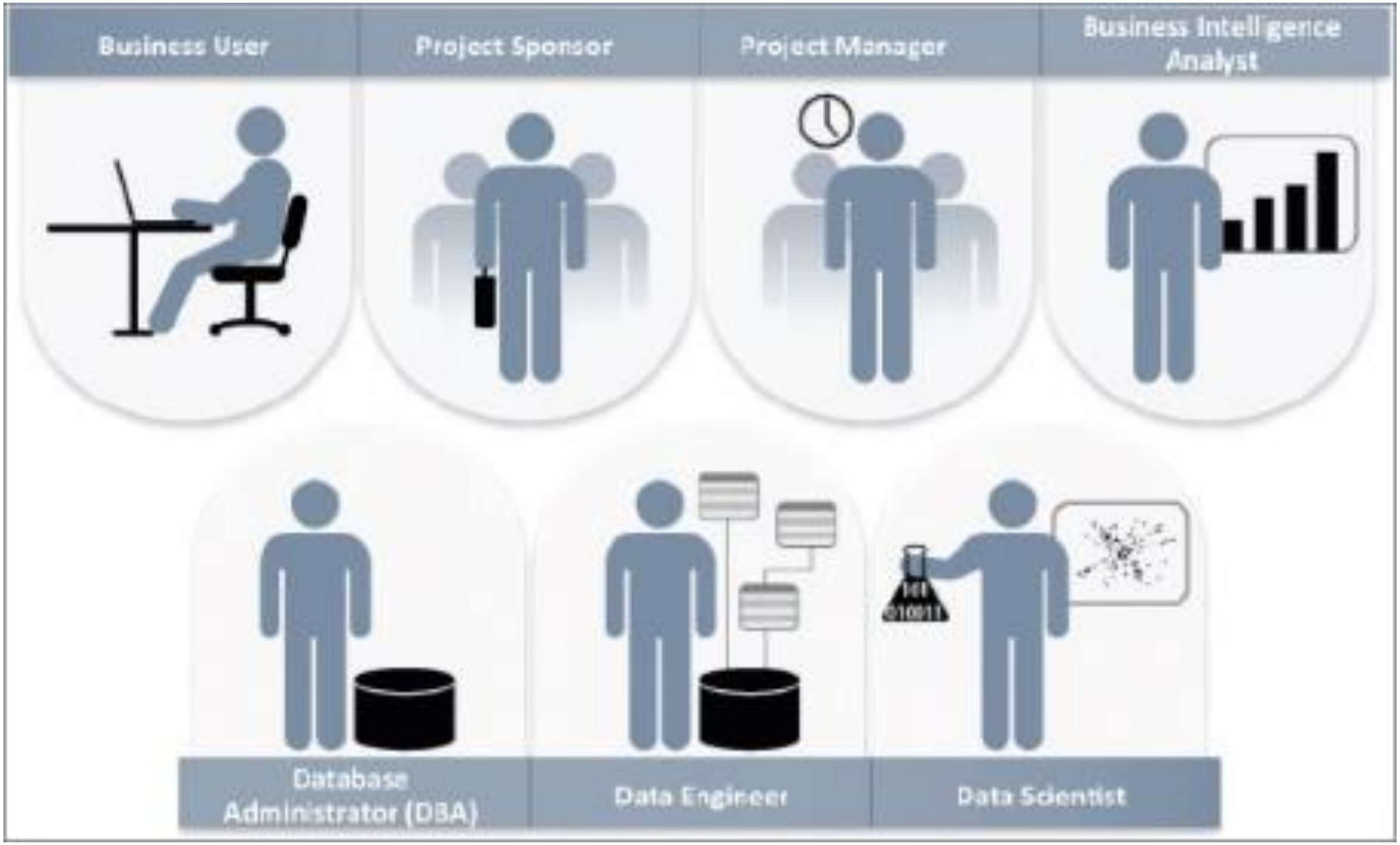
Data Science enables prediction/forecasting



Data Analytics just stops at present the most.

Data Science looks into the future.

Different stakeholders involved in a DS project



- **Business User**
  - Someone who understands the domain area and usually benefits from the results.
  - This person can advise the project team on the context of the project, the value of the results, and how the outputs will be operationalized.
  - Usually a business analyst, line manager, or deep subject matter expert in the project domain fulfills this role.

- **Project Sponsor**
  - Responsible for the genesis of the project.
  - Provides the impetus and requirements for the project and defines the core business problem.
  - Generally provides the funding and gauges the degree of value from the final outputs of the working team.
  - This person sets the priorities for the project and clarifies the desired outputs.

- **Project Manager**
  - Ensures that key milestones and objectives are met on time and at the expected quality.

- **Business Intelligence Analyst**
  - Provides business domain expertise based on a deep understanding of the data, key performance indicators (KPIs), key metrics, and business intelligence from a reporting perspective.
  - Generally create dashboards and reports and have knowledge of the data feeds and sources.

- **Database Administrator (DBA)**
  - Provisions and configures the database environment to support the analytics needs of the working team.
  - These responsibilities may include providing access to key databases or tables and ensuring the appropriate security levels are in place related to the data repositories.

- **Data Engineer**

  - Leverages deep technical skills to assist with tuning SQL queries for data management and data extraction, and provides support for data ingestion into the analytic sandbox.
  - Whereas the DBA sets up and configures the databases to be used, the data engineer executes the actual data extractions and performs substantial data manipulation to facilitate the analytics.
  - Works closely with the data scientist to help shape data in the right ways for analyses.

- **Data Scientist**
  - Provides subject matter expertise for analytical techniques, data modelling, and applying valid analytical techniques to given business problems.
  - Ensures overall analytics objectives are met.
  - Designs and executes analytical methods and approaches with the data available to the project.


WE
DATA
SCIENCE